TreeChecker: Screening Commodity Supply Chains for EUDR Deforestation Risk

10 October 2026, Oliver Webster-Mikkelsen

The problem

From 30 December 2026, large and medium EU importers of cattle, cocoa, coffee, palm oil, rubber, soy and wood must prove their goods come from land not deforested after 31 December 2020. Micro and small operators follow on 30 June 2027.

Every shipment needs the geolocation of the plots it came from. For an importer buying from thousands of smallholders, checking each plot against satellite data by hand is slow, technical and expensive.

Non-compliance is expensive. Each member state sets its own penalties, but the maximum fine must be at least 4% of the company's annual EU turnover. Goods and revenue can also be confiscated.

What TreeChecker does

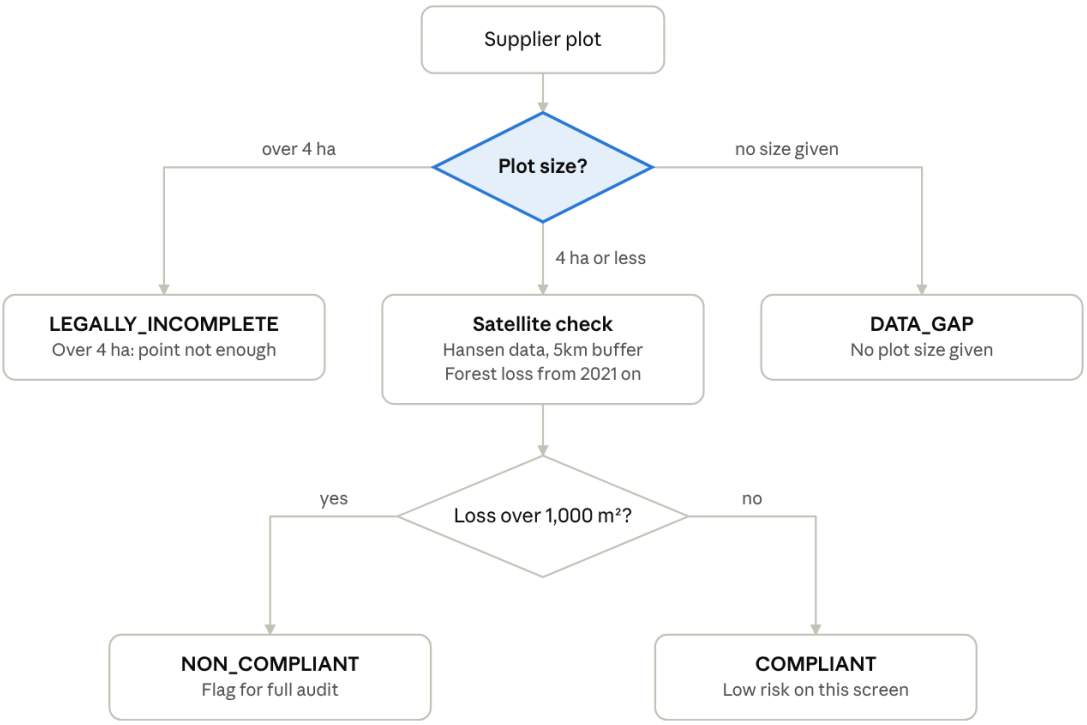
TreeChecker screens a whole supplier book for EUDR deforestation risk in one run, and ranks which suppliers to investigate first.

- **Input:** a spreadsheet of supplier plots, with coordinates, size in hectares, country, and, optionally, the importer's EU turnover.
- **Output per plot:** a compliance verdict, a risk score out of 100, an estimated financial exposure, and a PDF certificate with a satellite map as evidence.
- **Purpose:** triage. It flags where a full plot-level audit is needed. It does not give a final legal verdict.

How it works

TreeChecker applies EUDR's legal rules before it looks at any satellite data. This is the main design decision. EUDR requires full plot boundaries (polygons) above 4 hectares, so a point coordinate there is legally insufficient whatever the satellite shows.

Legal rules come first; only plots of 4 ha or less go to the satellite check



The satellite check uses Google Earth Engine and the Hansen Global Forest Change dataset. It sums forest loss since 2021 within 5km of the plot, because EUDR's cut-off is 31 December 2020.

Every verdict then feeds two numbers:

- **Risk score (0 to 100):** combines the verdict, the plot size and the country's risk tier (taken from the EU's official EUDR country benchmarking), so a whole supplier book can be ranked by priority.
- **Exposure estimate:** an illustrative worst case, expressed as a share of the importer's EU turnover and converted to euros only when turnover is supplied. Non-compliant plots use 4%, the level every member state's maximum fine must at least reach. Legally incomplete plots use 2% and plots with missing size data use 1%; both are my own assumptions.

Each plot also gets a PDF certificate with a satellite map showing tree cover and forest loss since 2021.

What I learned from the market

The one substantive response I got moved the problem upstream: firms often lack the plot data that screening depends on.

I contacted about 20 EUDR consultants and commodity importers on LinkedIn. I framed the messages as research into their compliance workflows, not as a sales pitch. One person replied: the Europe climate and regenerative agriculture lead at Proforest, who had worked on EUDR. His view was that the main

challenge is getting farm polygon data, because companies lack the traceability needed for most tropical commodities.

That changed how I see the opportunity. A screening tool is only as good as its input data, and for many smallholder supply chains that data does not exist yet. Traceability, meaning reliable plot boundaries for every supplier, looks like the bigger problem to solve.

The outreach also taught me something about method. Cold messages to senior people had a low response rate. Next time I would use warm introductions, trade association events and industry webinars, and follow up on promised referrals.

Limitations and next steps

TreeChecker is a screening signal, not legal proof. That role is deliberate; the buffer size needs recalibrating.

Limitations

- **Point, not plot.** A 5km buffer around a point measures forest loss across the surrounding landscape, not strictly inside the plot boundary.
- **Buffer too broad.** The current 5km buffer covers about 2,000 times the area of a 4 ha plot, and the 1,000 m² threshold is roughly one satellite pixel, so most plots in active forest landscapes will be flagged. Over-flagging is the right bias for triage, but at this scale it limits how useful the screen is.
- **Data coverage.** Hansen data is weaker in boreal regions, so a zero-loss reading there may reflect a gap in the data, not genuine compliance.
- **Illustrative fines.** The exposure figures are worst-case estimates, not predictions of actual penalties.

What I would build next

1. Size the buffer to the plot: a circle matching the declared plot area (about 113m radius for 4 ha), reported separately from a smaller landscape ring, with a threshold set as a share of area.
2. An auditability layer recording the data source, date range, confidence level and method behind each verdict.
3. Risk bands (Low / Medium / High / Very High) in place of a pass/fail verdict.
4. Portfolio analysis for 100 to 5,000 farms: share compliant, highest-risk locations, total exposure.
5. Longer term, the polygon data gap: helping firms build and verify plot boundaries for their suppliers.

Who to target first

Mid-sized importers whose supply base is mostly smallholder farms under 4 hectares, as in much of coffee and cocoa. There, EUDR accepts a point coordinate, which is exactly where TreeChecker's method is legally valid. Mid-sized firms must comply from December 2026 and are less likely to afford large consultancies. The catch is that smallholder chains also have the weakest traceability data, so some importers may not have a coordinate for every farm.